Activity Recognition Using Cell Phones

Machine Learning Mini-Project Report | UE24CS352A

1. Problem Statement

Human Activity Recognition (HAR) uses sensor measurements to identify the physical activity a person is performing. Smartphones contain accelerometers and gyroscopes that capture information about movement and orientation. The objective of this project is to build and evaluate machine-learning models that classify six activities: walking, walking upstairs, walking downstairs, sitting, standing, and laying down.

2. Dataset Description

This project uses the UCI Human Activity Recognition Using Smartphones dataset. It was collected from 30 volunteers who performed six activities while wearing a smartphone at the waist. The provided feature files contain 561 engineered features derived from inertial sensor signals sampled at 50 Hz and divided into fixed windows. The official training and test sets are retained.

Property	Value
Total examples	10,299
Training / test examples	7,352 / 2,947
Input features	561
Activity classes	6
Missing values detected	0 in training and test features

3. Methodology and Implementation

The workflow includes dataset loading and verification, training supervised classifiers, evaluating predictions on the provided test set, visualizing errors with a confusion matrix, and comparing model results. The project is implemented in Python using NumPy, pandas, scikit-learn, Matplotlib, Seaborn, and Joblib.

- **SVM (RBF kernel):** learns nonlinear decision boundaries for activity classification.
- **Random Forest:** combines predictions from multiple decision trees.
- **K-Nearest Neighbors (KNN):** predicts an activity from nearby training examples.
- **Feature selection:** SelectKBest with the ANOVA F-statistic selects 30 features; selection is fitted on training data only.

4. Experimental Results

Model	Features	Test accuracy	Weighted F1
SVM (RBF)	561	96.20%	96.19%
Random Forest	561	92.57%	92.55%
KNN (k=5)	561	90.16%	90.07%
SVM with 30 selected features	30	88.16%	88.09%

The full-feature RBF SVM achieved the highest test accuracy among the evaluated models. The confusion matrix shows strong performance on LAYING and notable confusion between SITTING and STANDING: 50 sitting examples were predicted as standing, while 10 standing examples were predicted as sitting. The 30-feature experiment reduced dimensionality from 561 to 30 features (about 94.65%) but accuracy decreased from 96.20% to 88.16%. This experiment also applies standardization, so the difference cannot be attributed to feature count alone.

5. Conclusion

The experiments demonstrate that smartphone sensor features can be used for human activity recognition. The RBF SVM performed best among the tested models at 96.20% accuracy, with Random Forest and KNN serving as comparison baselines. The feature-selection experiment illustrates a trade-off between lower dimensionality and predictive performance. Future work could examine alternative feature-selection methods, validation-based hyperparameter tuning, and generalization across subjects.

6. References

Reyes-Ortiz, J. L., et al. *Human Activity Recognition Using Smartphones*. UCI Machine Learning Repository. https://archive.ics.uci.edu/dataset/240/human%2Bactivity%2Brecogni-tion%2Busing%2Bsmartphones

Geis, T., Srinivas, S., and Talreja, R. *CS229 Final Report: Activity Recognition Using Cell Phones*, 2015.